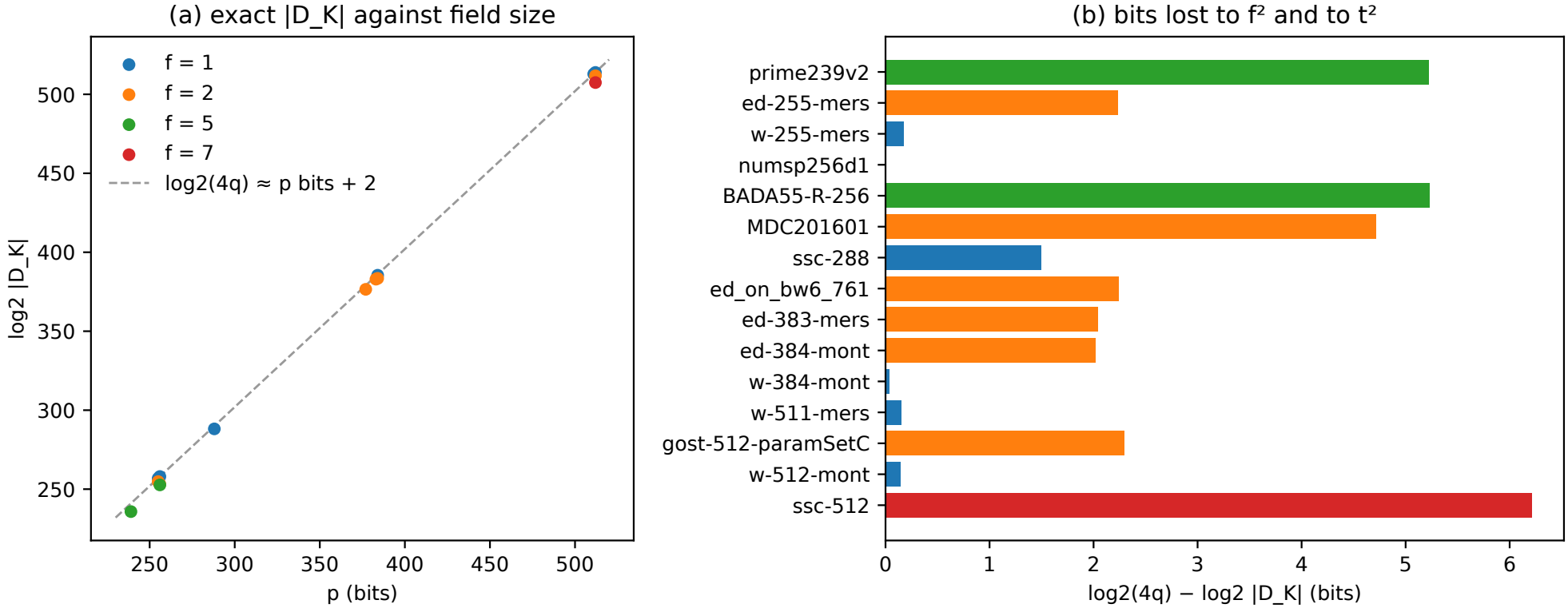
Exact CM discriminants of 18 standardized curves

research/endosweep_dk_exact_20261016 (2026-10-16). Data: `dk_exact.json`, `factorizations.jsonl`; std-curves commit `77fe6e3585ca2c2225b59d7df24b7c775437276f`. Every factor of $4q - t^2$ is proved prime by PARI `isprime`. These are exact computations, not measurements; the README states method, scope and regeneration. No D_K here was checked against the literature; seven agree with the `cm_disc` std-curves records.



n = 15 curves; exact integers (PARI isprime-proved factorisations), no uncertainty. Not shown: D_K = −3 on bn/bn510, bls/BLS12-638, arkworks/bw6_767.

Figure 1: (a) $\log_2|D_K|$ against p for the 15 curves without CM by construction, coloured by the conductor f. (b) The bits between $\log_2 4q$ and $\log_2|D_K|$. Recorded values from `dk_exact.json`.

Results

curve	p bits	4q−t² digits	s digits	f	D_K	log2 |D_K|	min degree	std-curves
x962/prime239v2	239	73	71	5	(71 digits)	235.78	2^233.78	cm_disc and conductor: agrees
nums/ed-255-mers	255	78	77	2	(77 digits)	254.77	2^252.77	no cm_disc
nums/w-255-mers	255	78	78	1	(78 digits)	256.83	2^254.83	cm_disc and conductor: agrees
nums/numsp256d1	256	78	78	1	(78 digits)	258	2^256	cm_disc and conductor: agrees
other/BADA55-R-256	256	78	77	5	(77 digits)	252.69	2^250.69	cm_disc and conductor: agrees
other/MDC201601	256	77	77	2	(77 digits)	253.2	2^251.2	no cm_disc
other/ssc-288	288	87	87	1	(87 digits)	288.16	2^286.16	cm_disc and conductor: agrees
arkworks/ed_on_bw6_761	377	114	113	2	(114 digits)	376.51	2^374.51	no cm_disc
nums/ed-383-mers	383	116	116	2	(116 digits)	382.96	2^380.96	no cm_disc
nums/ed-384-mont	384	117	116	2	(116 digits)	383.45	2^381.45	no cm_disc
nums/w-384-mont	384	117	117	1	(117 digits)	385.43	2^383.43	no cm_disc
bn/bn510	510	154	1	(77 digits)	-3	1.58	1	cm_disc and conductor: agrees
nums/w-511-mers	511	155	155	1	(155 digits)	512.85	2^510.85	no cm_disc
gost/id-tc26-gost-3410-2012-512-paramSetC	512	155	155	2	(155 digits)	511.71	2^509.71	no cm_disc
nums/w-512-mont	512	155	155	1	(155 digits)	513.85	2^511.85	no cm_disc
other/ssc-512	512	155	153	7	(153 digits)	507.44	2^505.44	no cm_disc
bls/BLS12-638	638	193	1	(97 digits)	-3	1.58	1	cm_disc and conductor: agrees
arkworks/bw6_767	767	232	1	(116 digits)	-3	1.58	1	no cm_disc

Excluded (already in research/endosweep_nist_20261006/factorizations.jsonl): P-256, P-384, P-521, Curve25519. Lean: `CryptoResearch.Endosweep.DkExact.exact_discriminants` (target `TASK-20261016-f6a5e8`) checks, per curve, $q+1-t=hn$, $t^2-4q=D_K f^2$, the product of the factors, the squarefree part and the D_K construction, $D_K \equiv 0,1 \pmod 4$ and the degree bound; fundamentality is conditional on the primality of the factors above 10^8 (PARI, not Lean).